

The unconditional block mean

one lemma: $\kappa^\dagger = 1\{q^\dagger > 1\}$;
the mean bound stays open

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Abstract

r440 showed that the block mean of $\kappa^\dagger = \text{ind}_-(R^\dagger - \frac{1}{2}I)$ is the winding of an averaged integrand, and that a fixed contour about $(0, 1)$ sees none of $\kappa^\dagger$ (the $1/2$ -cluster sits in a collar of $s = 1$). Combined with r433, that collar correction is the last Redheffer pivot: $\kappa^\dagger = 1\{q^\dagger > 1\}$. The block mean of $\kappa^\dagger$ is therefore the frequency of $q^\dagger > 1$ on the block. This is the one lemma of the round. An unconditional bound keeping that frequency strictly below 1 is *not* claimed: the unsigned Chebyshev/Mertens majorant already exceeds 1, and a signed-oscillatory bound would sit at the circularity gate. No RH claim.

Status. Lemma 1 is **SATZ**. Propositions 1–2 are machine-checked identities or named refutations. Proposition 3 is a sealed census. The unconditional frequency bound is *open*. Ausgang REDUZIERT / DICTIONARY_EXACT / UNSIGNED_MAJORANT_REFUTED / SELECTED_POINTWISE_CENSUS / MEAN_BOUND_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

r440 T1: $\kappa^\dagger = \#\{s \in (0, 1) : \tau^\dagger(s) = 0\}$ with $\tau^\dagger(s) = \det(I - sG^\dagger)$, $G^\dagger = (\mathfrak{T}^\dagger)^*\mathfrak{T}^\dagger$. A circle of radius 0.40 about $1/2$ winds 0 on every living window measured — the interior $(0.10, 0.90)$ is empty. So $\kappa^\dagger$ is entirely the collar correction: which side of $s = 1$ the one collar zero occupies. r433 identifies that side with the last Redheffer pivot $\delta = 1 - q^\dagger = -\text{sch}$.

Lemma 1 (Block-mean dictionary). *On the r409/r433 construction, $\kappa^\dagger = 1\{q^\dagger > 1\}$. Hence on any finite block,*

$$\frac{1}{K} \sum_k \kappa_k^\dagger = \frac{1}{K} \sum_k 1\{q_k^\dagger > 1\}.$$

Proof. Spectral calculus plus the last-pivot identity. Over $\mathbb{Q}$: $\delta = 187/450$, $q^\dagger = 263/450 < 1$, $1 - q^\dagger = -\text{sch}$. On w9 the three counters ($\text{ind}_-(R^\dagger - \frac{1}{2}I)$, zeros of $\tau^\dagger$ in $(0, 1)$, $1\{q^\dagger > 1\}$) agree at 0, with $|1 - q^\dagger + \text{sch}| = 9.5 \cdot 10^{-13}$. On selected $k = 3, 4, 5, 6, 7, 9$ they agree at 0; on dead χ (six windows) they agree at 1. $\square$

r430 `exists_index_zero_of_block_mean_lt_one` then turns $\liminf$ of the right-hand side < 1 into an index-zero window. The remaining analytic task is a bound on the frequency, not a new identity.

2 Source form of $q^\dagger$

r420/r424 write $q^\dagger = \gamma + v^T(I - BB^T)^{-1}v$ with $\gamma = \|b\|^2/B_w$ and b the μ -OP coefficient of the signed border measure σ . Woodbury gives the compact form $q^\dagger = B_w^{-1}b^T(I - C)^{-1}b$, $C = B^TB$.

Proposition 1 (Double sum).

$$q^\dagger = \sum_{a,a'} \frac{\sigma_a \sigma_{a'}}{B_w} \langle \phi(t_a), (I - C)^{-1} \phi(t_{a'}) \rangle.$$

On w9 the two reconstructions match $q^\dagger$ at $1.2 \cdot 10^{-15}$.

Selected anchors $a_k = 2^k$ are ζ -windows: the atoms a include primes in dyadic ranges. The block-averaged $q^\dagger$ is therefore an average of these finite double sums. Markov's inequality would convert $\frac{1}{K} \sum q_k^\dagger < 1$ into a positive density of $q^\dagger < 1$, hence into frequency < 1 . That implication is trivial; the missing input is the bound on the mean.

3 Gate: unsigned majorants fail

Proposition 2 (Unsigned envelope). *Replacing b by $|b|$ on w9 yields $38.49 \gg 1$. Replacing σ by $|\sigma|$ yields 2.10 already at $k = 5$ ($k_z = 17$) and 87.1 at $k = 9$ ($k_z = 116$).*

A positive Chebyshev/Mertens majorant on the dyadic prime mass therefore cannot prove $q^\dagger < 1$. The remaining bound needs the *signed* pairing of σ against $(I - C)^{-1}$. PNT controls the main term of Λ but not that pairing. An RH-quality error term on the oscillatory sum is the circularity gate (forbidden as a mean-value carrier). The unconditional-positive route is closed.

4 Census and trend

Proposition 3 (Selected liveness). *Selected $k = 3, 4, 5, 6, 7, 9$: $q^\dagger \in [0.891, 0.962]$, all < 1 , mean 0.932, $\kappa^\dagger = 0$ on 6/6 ($k = 8$ not rebuilt). The sequence is not falling: dip at $k = 5$ then rise toward 1 ($k = 9$: 0.962). Markov room 0.068. EXT-6: all $q^\dagger < 1$, mean 0.964. Core-42: 42/42 $q^\dagger < 1$, mean 0.961, min = 0.908 ($k_z 22$), max = 0.9985 ($k_z 16$); four windows sit above 0.99 and stay living. Dead χ 6/6: $q^\dagger > 1$ and $\kappa^\dagger = 1$. Living χ_3 -9: $q^\dagger = 0.935$, $\kappa^\dagger = 0$. No dead Selected window is known.*

The selected sequence is a ζ -window ladder. r428 already recorded that the old ladder-criticality was a quantifier artefact; the present census is compatible with Selected being pointwise living on the measured range. The honest risk is cofinal approach of $q^\dagger$ to 1 from below (vacuous MAIN $k_z 16$ sits at 0.9985), not a known crossing.

5 Kills

- **Soft mutants.** $\text{tr } G^\dagger$ is 50.9 on living w9 and 53.0 on dead χ_3 -15; $M_2 = 16 \text{tr}((I - R^\dagger)^4)$ is 32.96 vs 39.01. Both $\gg$ the threshold (r398, r440).
- **MI2.** On the block $\{w9, k_z 5\}$ the averaged-integrand winding equals the mean of windings at $4 \cdot 10^{-15}$, and the fixed $r = 0.40$ contour winds 0 (the collar carries all of $\kappa^\dagger$).
- **Scramble / circularity.** Unaugmented $\text{nneg}(R) = 21$. A pointwise bound $|\sum \Lambda n^{-1/2-it}| \ll X^\varepsilon$ is not a mean-value carrier.

Claim boundary

Nothing in this note is evidence for or against the Riemann hypothesis, in either direction. Lemma 1 is finite linear algebra. The frequency bound that would feed r430 is open.

References

- [1] Round 398, `high_moment_inertia`.
- [2] Round 420, `cj_sigma`: $\text{den} = 1 + \gamma - v^T s$.
- [3] Round 424, `gamma_chain`: b is the μ -Parseval coefficient of σ .
- [4] Round 428, `qn_reopened`: ladder criticality as a quantifier artefact.
- [5] `exists_index_zero_of_block_mean_lt_one`.
- [6] Round 433, `edge_redheffer`: $\delta = 1 - q^\dagger = -\text{sch}$.
- [7] Round 440, `mean_tau_index`: T1/T2/MI2 and the collar anatomy.